



Improving approximate neural networks for perception tasks through specialized optimization

Cecilia De la Parra^{a,*}, Andre Guntoro^a, Akash Kumar^b

^a Robert Bosch GmbH, 71272 Renningen, Germany

^b Chair of Processor Design TU Dresden 01069 Dresden, Germany

ARTICLE INFO

Article history:

Received 13 January 2020

Received in revised form 11 June 2020

Accepted 13 July 2020

Available online 22 July 2020

Keywords:

Approximate neural networks

Approximate computing

Approximate multipliers

Neural network optimization

ABSTRACT

Approximate Computing has been proven successful in reducing the energy consumption of Deep Neural Networks (DNNs) implemented in embedded systems. For efficient DNN approximation at software and hardware levels, a specialized simulation environment and optimization methodology are required, to reduce execution and optimization times, as well as to maximize energy savings. Traditional frameworks for cross-layer approximate computation of DNNs are generally built only for simulation of convolutional and fully-connected layers, limiting the DNN types to be optimized through approximations. In this work, we present a specialized simulation environment for approximate DNNs, which allows for optimization of several DNN architectures built with more complex DNN layers such as depthwise convolutions and Recurrent Neural Units (RNNs) for time series processing. Low execution time overhead is achieved hereby through efficient GPU acceleration. Additionally, we deliver an analysis of approximate DNN and RNN robustness against quantization noise and different approximation levels. Finally, through specialized approximate retraining, we achieve promising energy savings and negligible accuracy losses with highly complex DNNs for image classification with ImageNet, such as MobileNet, and RNNs for keyword spotting with the Speech Commands Dataset.

© 2020 Elsevier B.V. All rights reserved.

1. Introduction

In recent years, several paradigms to optimize the hardware resources of deep learning applications have been proposed. One promising approach is *cross-layer approximate computing*, which involves the combination of approximation techniques at software and hardware level to reduce computational resources in embedded applications.

Approximations at a software level, such as pruning or quantization, have been thoroughly investigated even for highly complex DNNs dedicated to real-life tasks, such as large-scale image recognition or semantic segmentation for automated driving [1]. This has been possible thanks to specialized open source machine-learning frameworks such as Tensorflow Lite [2], or Ristretto [3], which allow a fast exploration of software-oriented approximation techniques.

On the other hand, approximations at hardware level are more difficult to explore, evaluate and optimize, mainly because the simulation of Approximate Computational Units (ACUs) largely increases the runtime in traditional frameworks for approximate computing. Furthermore, specialized cross-layer simulation

frameworks for approximate DNNs are generally implemented for CPU, which also increases training and validation times, when compared to exact GPU implementations [4]. In recent works, GPU-based simulation frameworks for cross-layer approximation, such as *Concrete* [5] and *ProxSim* [6], have been proposed to accelerate the optimization of approximate DNN architectures with common neural layers such as convolutional and fully-connected layers. However, more complex operations such as depthwise convolutions [7] or building blocks for Recurrent Neural Networks (RNNs) [8] are not available, restricting the architectures that can be approximated in such specialized frameworks.

In this work, we update *ProxSim*, presented in [6], by implementing more complex approximate layers, including depthwise convolution, Hadamard product and Gated Recurrent Units (GRUs), as well as more specialized methods for DNN quantization. This variety of layers and quantization methodologies allows us not only to optimize a wide range of approximate DNNs for different applications such as image recognition or keyword spotting, but also to evaluate the accuracy and robustness of such DNNs against quantization noise at different levels.

In summary, we make the following contributions:

- *ProxSimV2*: A specialized simulation framework for cross-layer DNN approximations such as low bitwidth quantization and approximate hardware, in a variety of layers

* Corresponding author.

E-mail address: cecilia.delaparra@de.bosch.com (C. De la Parra).